# Weekly Engineering Log – Week 7

**Date:** October 15, 2026

**Milestone target:**

Advance the PC cooling software prototype by testing the cooling decision logic, improving PC data logging, and preparing for hardware integration.

**Planned work:**

- Expand the Python monitoring application.
- Create and run unit tests for the cooling decision logic.
- Record PC performance data for later analysis.
- Determine the remaining hardware requirements and place the parts order.
- Establish weekly deliverables for the remainder of the project.

**Completed work:**

- Continued developing the Python application using psutil and GPUtil.
- Separated PC data collection and cooling decision logic into individual functions.
- Implemented initial decision logic that maps GPU temperature readings to simulated fan commands.
- Created a unit test file to check normal, warm, hot, critical, and unavailable GPU temperature conditions.
- Successfully ran the unit tests to verify the decision logic independently of the physical hardware.
- Placed the parts order for hardware.

**Technical evidence:**

- cooling_monitor.py containing the monitoring and decision logic.
- test_decision.py containing five unit tests.
- Terminal screenshot showing all five tests passing.
- Terminal output from the monitoring application.
- CSV log of collected PC performance data, once generated.

**Problems encountered:**

The hardware portion remains behind schedule because the microcontroller and supporting electronics have not yet been obtained. The software can currently simulate fan commands, but it

cannot control the physical fans. CPU temperature also remains unresolved because the current monitoring setup provides CPU utilization rather than a direct CPU temperature measurement.

**Hypotheses/troubleshooting:**

Testing the control logic with simulated temperature values allows the decision-making portion to be verified before hardware integration. The next step is to confirm that real PC data can be collected and logged consistently. Once the hardware is obtained, the software's simulated command function can be replaced with a communication interface to the microcontroller.

**Engineering decisions:**

Continue developing and testing the software independently while obtaining the hardware. Delay selecting the final fan driver and power supply until the fan specifications and microcontroller are confirmed. Keep enclosure fabrication as a lower priority until the control circuit and fan placement have been tested.

**Schedule status: Green.**

The software testing milestone has progressed, with software and PC communication working as intended. Since the parts are arriving later than expected, code advancements were made to stay on schedule, with hardware integrations to be focused on in the next two weeks. Meanwhile, I will continue software development while waiting for delivery, and prioritize basic hardware communication and fan control before beginning the main cooling experiments.

**Next actions:**

- Verify that the monitoring application produces consistent readings and CSV output.
- Start putting physical hardware together.
- Identify the fan connector, voltage, and current requirements.
- Establish communication between the PC software and microcontroller.
- Demonstrate at least two fan-control states.
- Record baseline PC performance and temperature measurements.
- Compare baseline results against results with the supplemental cooling system active.
- Use the measurements to select a fan placement or control configuration and document the final results.

Revised Schedule:

| Week | Target | Required deliverable |
|---|---|---|
| 8 | Hardware procurement | Confirmed parts, assembly plan, and updated delivery dates |
| 9 | Hardware bring-up | Microcontroller communicates with PC software; basic fan-control demonstration |
| 10 | Baseline testing | Recorded PC performance and temperature data without supplemental cooling |
| 11 | Cooling experiments | Comparable measurements with supplemental cooling enabled |
| 12 | Configuration adjustment | Compare fan placement/configurations and select a promising setup |
| 13 | Validation | Repeat tests, analyze results, and document limitations |
| 14 | Final submission | Final demonstration, results, and project documentation |

Terminal Output:

```
PS C:\Users\jtros\Documents\Projects\PC Cooling System> python -m unittest -v test_decision
test_critical_temperature (test_decision.CoolingDecisionTests.test_critical_temperature) ... ok
test_hot_temperature (test_decision.CoolingDecisionTests.test_hot_temperature) ... ok
test_missing_temperature (test_decision.CoolingDecisionTests.test_missing_temperature) ... ok
test_normal_temperature (test_decision.CoolingDecisionTests.test_normal_temperature) ... ok
test_warm_temperature (test_decision.CoolingDecisionTests.test_warm_temperature) ... ok

----------------------------------------------------------------------
Ran 5 tests in 0.001s
```